Bathymetric Reconstruction Using Time-Series Satellite Images and Minimal In Situ Data

Hongjun Youn and Taejung Kim

***Abstract*—This study proposes a method for reconstructing bathymetric terrain heights using time-series satellite images combined with a bathymetric interpolation technique. Conventional approaches of bathymetric surveys require LiDAR or echo sounders, which require significant cost and time. To overcome this limitation, we propose an algorithm to reconstruct an internal digital elevation model (DEM) of terrains under water bodies. We extract water boundaries from time-series satellite images. The extracted boundaries are treated as contour lines of bathymetric terrain, whose elevation values are not known. The contour lines are used as constraints during a hydrological bathymetric interpolation process. By assigning a true elevation to one water boundary and utilizing elevation values of nearby terrain, we reconstruct a bathymetric DEM. We can estimate a new water level from a new satellite image by overlaying a new water boundary on the bathymetric DEM reconstructed. We can validate the bathymetric DEM by comparing the estimated water level with the reference in situ water level. The proposed method was tested on three reservoirs using high-resolution KOMPSAT-3A images. With one reference water boundary and five nonreference water boundaries, the proposed method achieved an overall mean absolute error (MAE) of 0.61 m using six validation images for each reservoir, with MAEs of 0.61, 0.75, and 0.47 m for Nogok, Maegok, and Pungdo, respectively. These results show that time-series water boundaries can serve as effective constraints for bathymetric reconstruction and water-level estimation in small- and medium-sized reservoirs with limited in situ measurements.**

***Index Terms*—Australian National University Digital Elevation Model (ANUDEM) interpolation, bathymetry reconstruction, time-series satellite imagery, water-level estimation.**

I. Introduction

RESERVOIR water level and water capacity are critical components of water resource management, supporting flood control, drought mitigation, and water allocation [1], [2]. In the absence of precise bathymetric terrain models or continuous in situ observations, it becomes difficult to monitor reservoir water levels effectively [3], [4], [5]. This may lead to operational gaps that hinder timely disaster response and management planning [6]. In Korea, water levels of medium- to large-scale reservoirs with effective storage capacities exceeding 100 000 tons are managed and monitored by the Korea Rural Community Corporation. However, smaller reservoirs under the jurisdiction of local governments often lack sufficient staffing and equipment for routine water-level monitoring [7], [8]. Bathymetric surveys such as LiDAR and echo sounders offer reliable bathymetric terrain models. However, they require significant cost and time [9], [10], [11], [12].

To overcome these limitations, satellite images have garnered increasing attention in the field of hydrology [12], [13]. Satellite images offer repeated, wide-area coverage and are effective for monitoring water resources over inaccessible regions [8], [14], [15]. Satellite-based methods for estimating water levels can be categorized into methods using radar altimeters, synthetic aperture radar (SAR) images, and optical images [16].

Radar altimeters enable the observation of inland water-level variations with decimeter-level accuracy [17], [18], [19]. They have been used to monitor major lakes worldwide via satellites such as ENVISAT, TOPEX/Poseidon, the Jason series, and Sentinel-3. However, their narrow ground-track coverage limits its applicability to small and medium-sized reservoirs unless a satellite pass intersects the waterbody [20], [21], [22], [23]. SAR images can capture data regardless of cloud cover or lighting conditions, but their effectiveness in delineating water boundaries is constrained by surface scattering effects and spatial resolution limitations [24], [25], [26]. In contrast, optical images can estimate water level changes by identifying the water surface using multispectral analysis. For example, Landsat and Sentinel-2 have been used to track surface area fluctuations in large reservoirs over long time periods, enabling estimation of seasonal or monthly storage variations [27], [28]. On the other hand, approaches using single optical satellite images for water monitoring face a fundamental limitation: a satellite image alone cannot directly provide the elevation of the water surface [29]. Auxiliary information such as a bathymetric digital elevation model (DEM) is required for accurate water-level estimation [30]. Since bathymetric surveys to build a precise DEM under waterbodies are very costly, various methods have incorporated existing topographic terrain DEMs [31]. For instance, height values corresponding to observed water extents were extracted from topographic DEMs [32], [33]. However, errors may arise due to DEM resolution, lack of vertical accuracy under waterbody, and topographic changes over time [34]. To overcome these challenges, recent studies have explored multitemporal approaches [35], [36]. These methods typically derive empirical relationships such as water level versus surface area by calculating water surface area

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from a sequence of satellite images and combining them with corresponding in situ measurements or existing elevation-area capacity curves [37], [38].

Recent satellite-derived bathymetry (SDB) research has increasingly used deep learning to estimate water depth from multispectral reflectance [39], [40]. Multiscale fusion models, transformer architectures, and physics-guided neural networks have been proposed for optical bathymetric inversion [41], [42], [43]. At the reservoir scale, deep learning has been applied to reconstruct area–storage–depth relationships from large reservoir datasets [4]. Where in situ sounding data are limited, airborne and photogrammetric data have been used to provide bathymetric reference data [44], [45]. Despite these advances, most optical SDB methods still require reliable reference depth data or a detectable bottom-reflectance signal. This requirement limits their use in small- to medium-sized inland reservoirs, where depth samples are sparse and bottom visibility is often reduced by turbidity, aquatic vegetation, and seasonal changes in water quality.

Unlike reflectance-based SDB approaches, the proposed method uses time-series water boundaries as geometric constraints for bathymetric reconstruction. Each boundary represents the intersection between an approximately level water surface and the reservoir terrain. As water levels rise or fall, the time-series boundaries provide relative water-level ordering according to their spatial hierarchy and are treated as contour-like constraints. One reference water boundary is assigned a true elevation value from an in situ water-level measurement, while nonreference water boundaries provide contour-like constraints without explicit elevation values. Together with nearby topographic DEM cells, these constraints are used in Australian National University Digital Elevation Model (ANUDEM)-based hydrological bathymetric interpolation. A new water level is estimated by averaging elevation values from the reconstructed bathymetric DEM along a newly extracted water boundary.

In this process, errors in water boundary extraction and in the elevation assigned to the reference water boundary can affect the reconstructed bathymetric DEM quality. A boundary extraction error shifts the spatial position of the corresponding constraint, whereas an error in the reference elevation propagates into the reconstructed bathymetric DEM. Although time-series water boundaries reduce reliance on a single extracted boundary, these errors can still affect the interpolation result. Therefore, the reconstructed DEM was evaluated by comparing water levels estimated from independent satellite images with the in situ water-level measurements.

Accordingly, this article proposes a boundary-constrained bathymetric reconstruction method for small- to medium-sized reservoirs where detailed bathymetric survey data and continuous water-level monitoring are not routinely available.

II. Methodology

Fig. 1 shows the bathymetric reconstruction method proposed here. The workflow is divided into three parts: water boundary extraction, hydrologic interpolation, and new water-level estimation. For water boundary extraction, time-series satellite images are used as inputs. They are used to extract water areas at different image acquisition times and thereby form water boundary contours. For hydrologic interpolation, a topographic DEM around the water area and one reference water boundary, the water boundary with a true height value, are required. They are used as constraints for hydrological bathymetric interpolation. Once a bathymetric terrain is constructed, a new water level can be estimated from a new single satellite image by fitting the new water boundary from the new image onto the bathymetric terrain. In this article, new water-level estimation is carried out to check the accuracy of bathymetric estimation as well as the validity of the proposed method for reservoir water-level estimation.

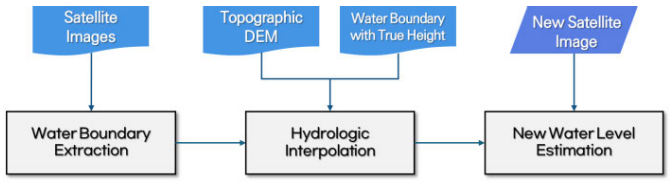


Fig. 1. Flowchart of water-level estimation.

A. Water Boundary Extraction

A water boundary represents the interface between land and water at a given water level. It implies that a terrain elevation along this boundary corresponds to the water level at that time. As water levels drop or rise, terrain areas with certain elevations emerge or submerge. Accordingly, water boundaries observed at different times can be interpreted as analogs to elevation contours. Note that water boundaries extracted in this stage contain 2-D contours whose explicit elevation values are not known. If true elevation values are available for some of these boundaries, they can be used as reference boundaries for bathymetric reconstruction.

Before normalized difference water index (NDWI) calculation, the time-series images for each reservoir were co-registered to a reservoir-specific reference image so that all water boundaries were extracted in a common spatial reference. This preprocessing step allowed boundaries from different acquisition dates to be used consistently as contour-like constraints for bathymetric reconstruction.

For each reservoir of interest, satellite images taken at multiple times are collected. The time-series images include a range of water-level conditions, from high- to low-water levels. The variation in water levels provides the spatial variation of water boundaries. The NDWI is used to extract water boundaries. NDWI is a widely adopted spectral index that distinguishes water bodies by calculating the reflectance difference between the green and near-infrared (NIR) bands [46]. Vegetation and soil exhibit high reflectance in the NIR band. Water strongly absorbs NIR and thus exhibits low reflectance in NIR bands and high reflectance in the green band. Based on these spectral properties, NDWI is defined as follows:

$$\mathbf{NDWI} = \frac{\mathbf{Green} - NIR}{Green + NIR}. \tag{1}$$

For each reservoir, a reservoir-centered region of interest (ROI) is selected on the image to include all observed water

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Fig. 2. Water boundary extraction using NDWI. (a) Original image. (b) NDWI mask. (c) Extracted water boundary.

extents and surrounding land areas, and the same ROI is applied to all acquisition dates. An NDWI histogram is then generated for the ROI, and Otsu's method is used to determine the threshold that maximizes the between-class variance between water and nonwater pixels. Pixels with NDWI values greater than or equal to the threshold are classified as water. The resulting water mask is converted into polygons, and the reservoir boundary is extracted from the corresponding water-mask polygon. The extracted boundaries are visually checked against the corresponding pan-sharpened optical images, and limited manual refinement is applied only where local water boundary errors caused by shadows, mixed pixels, or adjacent land-cover features are evident.

The final water boundaries represent the observed water extent at each acquisition time under the assumption of an approximately level water surface. Fig. 2(a) shows the original satellite image, Fig. 2(b) shows the NDWI-based water mask, and Fig. 2(c) shows the extracted water boundary overlaid on the base image.

B. Hydrological Interpolation

Hydrological bathymetric interpolation generates a DEM by integrating hydrologically significant features such as elevation points, sinks, and cliffs. It ensures hydrological consistency by avoiding artificial obstructions and unrealistic flow paths and forms continuous drainage networks that resemble natural terrain [47], [48], [49]. This study adopts ANUDEM [47], [48], [50] for hydrological bathymetric interpolation. ANUDEM is based on 2-D minimum-curvature interpolation and is designed to preserve input elevation data while minimizing surface curvature [47]. The algorithm employs finite difference methods to iteratively adjust curvature during DEM generation [50]. Hydrological constraints such as water boundaries and water boundaries with reference elevations can be incorporated during the ANUDEM interpolation process.

The water boundaries extracted from the previous stage are divided into two types. One is a reference water boundary assigned a true elevation value. The others are nonreference water boundaries without a true elevation value. Reference and nonreference water boundaries and a topographic DEM are provided as inputs for hydrological interpolation. For nonreference water boundaries, the relative order of water-surface elevation is inferred from their spatial hierarchy under the assumption that the water surface is approximately level. If the boundary at time A lies within that of time B, the boundary at time A is interpreted as representing a lower water level than the boundary at time B. These boundaries provide relative contour-like constraints on bathymetric interpolation. The surface elevations are iteratively adjusted during interpolation, and the areas between them are interpolated smoothly based on the minimum-curvature principle, enabling bathymetric reconstruction.

A reference water boundary imposes another constraint on bathymetric interpolation. It provides a true elevation contour and enables ANUDEM to generate a bathymetric DEM that better reflects hydrological realism. To minimize in situ data use, one absolute elevation is provided for a water boundary extracted during a low-water period. Through hydrological interpolation, the surrounding surface is adjusted to form a concave shape centered on this contour. This approach enables accurate bathymetric reconstruction with minimal water-level surveying.

Bathymetric DEM reconstruction is performed using the Topo to Raster (ANUDEM) tool in ArcGIS Pro. The reference water boundary is supplied as a contour input (TopoContour) with the observed water level as its elevation attribute; the nonreference water boundaries extracted from the time-series images are supplied as Lake inputs (TopoLake) without measured elevation attributes; and the surrounding topographic DEM is supplied as PointElevation inputs (TopoPointElevation). The output cell size is set to 0.55 m to match the pan-sharpened KOMPSAT-3A image grid.

The gridding configuration is selected through preliminary reconstruction trials. The trials are used to check whether the reconstructed DEMs preserve the basin shape constrained by the water boundaries, without excessive smoothing or local interpolation artifacts. The adopted configuration uses maximum iterations = 25, roughness penalty = 0, profile-curvature roughness penalty = 0, discretization error factor = 1, Tolerance 1 = 2.5, and Tolerance 2 = 100. Roughness and profile-curvature penalties are set to 0 to avoid additional roughness or profile-curvature weighting beyond the ANUDEM minimum-curvature formulation. Tolerance 1 controls the balance between elevation constraints and drainage enforcement, whereas Tolerance 2 limits drainage clearance across unrealistically high terrain barriers. The same configuration is applied to all reservoirs without reservoir-specific tuning to maintain comparable reconstruction conditions across different basin geometries.

As a result of interpolation, the generated bathymetric DEM embeds the multiple water boundaries as pseudo-contour constraints, and the reservoir's bathymetry is reconstructed based on the interpolated elevation between these boundaries.

C. New Water-Level Estimation

The satellite images used for water-level estimation are processed with the same ROI selection, image preprocessing, and image-to-image co-registration procedure before the water boundaries are extracted.

Based on the reconstructed bathymetric DEM, the reservoir water level at a specific time can be estimated using a new single satellite image. The process begins by extracting a new

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(a) (b) (c)

Fig. 3. Test reservoirs in this study. (a) Nogok reservoir. (b) Maegok reservoir. (c) Pungdo reservoir.

water boundary from the new image. Then, the new boundary is aligned with the bathymetric DEM. Elevation values of corresponding bathymetric DEM cells along that boundary are averaged and defined as the reservoir's new water level for that specific time.

Since water surfaces remain horizontally level due to gravitational effects, elevation values along the water boundary tend to be consistent despite minor local fluctuations. Therefore, the mean elevation of the boundary serves as a reliable quantitative estimate of the water level. This averaging approach also helps mitigate the effects of noise and outliers in the elevation data, contributing to more stable and robust water-level estimations.

In this study, the estimated water levels are compared to true in situ water levels for two purposes. First, the comparison aims to verify the validity of the proposed bathymetric reconstruction and the accuracy of the reconstructed bathymetric DEM. Second, it evaluates the feasibility of the water-level estimation method using the reconstructed bathymetric DEM. For the accuracy assessment, each of the six images other than the reference water boundary image is excluded in turn from bathymetric reconstruction and used for water-level estimation. For each configuration, the required number of reference and nonreference water boundaries is selected from the remaining images. The estimated water level is obtained by averaging the elevations of the reconstructed DEM cells along the excluded water boundary and is compared with the corresponding in situ water level. The estimation error is defined as the difference between the estimated water level and the corresponding in situ water level; therefore, a positive bias indicates overestimation. Bias, standard deviation (SD), mean absolute error (MAE), root-mean-square error (RMSE), and median absolute error are calculated from the estimation errors. The same validation procedure is also used for method comparison and uncertainty assessment. The comparison included direct topographic DEM sampling, IDW, spline, and empirical Bayesian kriging. The uncertainty assessment examined image co-registration error, water-boundary extraction error, and ANUDEM parameter settings.

III. Experiment Data

To verify the practical applicability and quantitative accuracy of the proposed water-level estimation method, experiments were conducted on three small-to-medium-sized reservoirs. Three small-to-medium reservoirs with varying geometries were selected as study sites, each exhibiting distinct topographic and hydrological characteristics. They are illustrated in Fig. 3. The first reservoir, Nogok in Gyeonggi Province, is situated in relatively flat terrain and features a simple structure, with gradual elevation changes along its boundaries. The second, Maegok at Jeollanam Province, exhibits a complex basin structure and presents a more complex topographic configuration than the other two sites. The third, Pungdo Reservoir at Jeollanam Province, has an irregular and asymmetric shape, with complex terrain distributed unevenly across the basin.

TABLE I
Specifications of Satellite Images Used in Experiments (Nogok Reservoir)

No.	Satellite	Date of acquisition	Water reserve rate (%)	Water level (m)
1	KOMPSAT-3A	2020. 04. 28	100.0	79.31
2		2021. 11. 27	99.9	79.30
3		2021. 02. 08	96.5	79.19
4		2020. 10. 20	85.0	78.79
5		2021. 09. 28	84.5	78.75
6		2021. 05. 13	72.1	78.34
7		2021. 07. 28	47.3	76.40

TABLE II
Specifications of Satellite Images Used in Experiments (Maegok Reservoir)

No.	Satellite	Date of acquisition	Water reserve rate (%)	Water level (m)
1	KOMPSAT-3A	2016. 12. 31	94.7	9.87
2		2016. 04. 15	94.5	9.84
3		2017. 04. 28	93.6	9.81
4		2017. 06. 02	52.0	7.62
5		2017. 11. 09	44.5	7.14
6		2017. 11. 06	40.5	6.83
7		2017. 09. 13	35.5	6.00

High-resolution KOMPSAT-3A Level-1G optical images were used in this study. The Level-1G products were geometrically corrected and projected to UTM by the data provider and included the radiometric and sensor corrections applied during Level-1R processing. The images included one panchromatic (PAN) band and four multispectral bands: red, green, blue, and NIR. Additional atmospheric correction was not performed beyond the product-level radiometric corrections. To reduce the sensitivity of water-mask extraction to interdate differences in image conditions, an Otsu threshold was estimated separately for each acquisition within the selected ROI.

After PAN-sharpening, the spatial resolution of the multispectral bands was improved to 0.55 m, and the pan-sharpened green and NIR bands were used for NDWI calculation. For each reservoir, seven time-series images were prepared, with at least one image capturing conditions where the reservoir storage was below 50%.

To reduce residual interdate offsets in the Level-1G images, feature-based image-to-image co-registration was performed within each ROI. One image was selected as the co-registration reference image for each reservoir, and all images used for

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Fig. 4. Representative time-series satellite images on Pungdo reservoir.

TABLE III
SPECIFICATIONS OF SATELLITE IMAGES USED IN EXPERIMENTS (PUNGDO RESERVOIR)

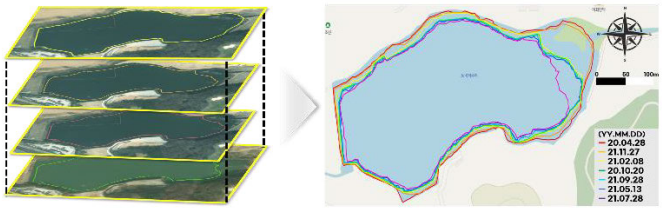
No.	Satellite	Date of acquisition	Water reserve rate (%)	Water level (m)
1	KOMPSAT-3A	2017. 04. 28	84.9	6.39
2		2017. 11. 06	82.8	6.15
3		2017. 11. 09	82.0	6.14
4		2016. 04. 15	72.2	5.70
5		2016. 12. 31	65.0	5.32
6		2017. 09. 13	64.1	5.28
7		2017. 06. 02	39.0	3.92

TABLE IV
IMAGE-TO-IMAGE CO-REGISTRATION ACCURACY OF KOMPSAT-3A IMAGES

Reservoir	Co-registration Reference date	Mean (px / m)	Max. (px / m)
Nogok	2021.07.28	0.9946 / 0.5470	1.1584 / 0.6371
Maegok	2017.11.09	1.1016 / 0.6059	1.2414 / 0.6828
Pungdo	2017.06.02	1.1333 / 0.6233	1.1858 / 0.6522

bathymetric reconstruction and validation were co-registered to this reference image. Tie points were extracted using the ORB detector and descriptor. Descriptor matching was performed using a brute-force matcher with k-nearest-neighbor matching; water surfaces and water boundaries were excluded because their positions change with water level. Outlier matches were removed using RANSAC, and an affine transformation was estimated for each nonreference scene. The co-registered images were resampled to the 0.55 m pan-sharpened grid of the reservoir-specific reference image using bilinear interpolation. Co-registration accuracy was assessed using independent checkpoints that were not used in the transformation estimation.

In situ water-level observations were obtained from the Rural Agricultural Water Resources Information System (RAWRIS), operated by the Korea Rural Community Corporation. Water-level observations corresponding to the satellite acquisition dates were used in the analysis. These measurements were referenced to elevation above mean sea level. The



(a) (b)

Fig. 5. Time-series water boundaries from Nogok reservoir. (a) Sequential KOMPSAT-3A images and (b) water boundaries.

acquisition dates and the corresponding water-level values are summarized in Tables I–III. Fig. 4 shows representative time-series KOMPSAT-3A satellite images of the Pungdo reservoir, showing a gradual decrease in the water level over acquisition dates. These images serve as reference data for subsequent boundary extraction.

The seven time-series images for each reservoir are listed in Tables I–III. One image acquired during a low-water period was used as the reference water boundary image. Each of the remaining six images was excluded in turn from bathymetric reconstruction and used for water-level estimation. The reference water boundary images corresponded to the lowest water levels listed in the tables: 2021. 07. 28 for Nogok, 2017. 09. 13 for Maegok, and 2017. 06. 02 for Pungdo.

For each reservoir, co-registration RMSE was calculated for each nonreference scene using independent checkpoints that were not used in the transformation estimation. The mean and maximum RMSE values across the co-registered scenes are reported in Table IV. Pixel RMSE values were converted to meters using the 0.55 m pan-sharpened pixel size.

IV. EXPERIMENT RESULTS

A. Water Boundary Extraction Results

Fig. 5 shows the time-series water boundaries extracted from the Nogok Reservoir. Fig. 5(a) shows sequential KOMPSAT-3A satellite images, from which water extents were delineated using NDWI, followed by contour extraction. Fig. 5(b) overlays the extracted water boundaries corresponding to different acquisition dates.

Fig. 6 shows the time-series water boundaries extracted from Nogok, Maegok, and Pungdo reservoirs. Each color represents a different acquisition date, indicating temporal variations in water extent. The water boundaries were derived

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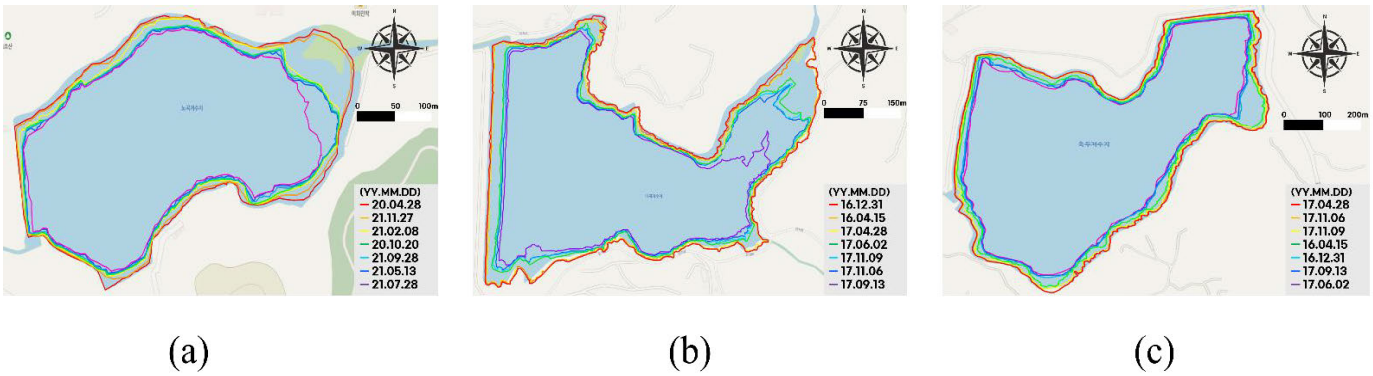


Fig. 6. Time-series water boundaries from (a) Nogok, (b) Maegok, and (c) Pungdo reservoirs.

TABLE V
OTSU THRESHOLDS FOR NDWI WATER-MASK GENERATION BY RESERVOIR

Reservoir	Mean threshold	Range (min / max)	SD
Nogok	-0.0654	-0.1609 / 0.0084	0.0966
Maegok	-0.0841	-0.1962 / 0.0065	0.0914
Pungdo	-0.0745	-0.1740 / 0.0063	0.0671

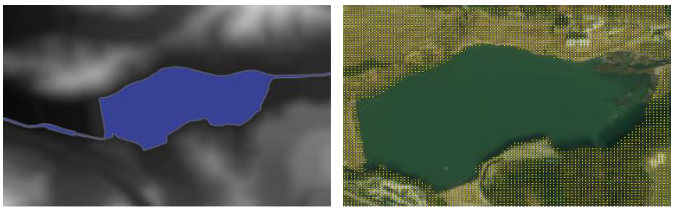


Fig. 7. Topographic DEM around the Nogok reservoir. (a) Reservoir and river sections on the DEM and (b) topographic DEM cells.

using the same NDWI-based water-mask generation and boundary extraction procedure. All boundaries were overlaid after image-to-image co-registration to the reservoir-specific reference image.

Table V summarizes the Otsu thresholds estimated from NDWI histograms within the selected ROI for each reservoir. The thresholds varied by reservoir and acquisition date, supporting the use of image-specific thresholding rather than a single fixed NDWI threshold.

The extracted boundaries represent temporal variations in water extent. Larger outer boundaries correspond to high-water conditions, while smaller inner boundaries reflect low-water levels. The spatial nesting of boundaries indicates the relative variation of water levels across the acquisition period.

The time-series water boundaries derived in this step served as key inputs for hydrological bathymetric interpolation. By acting as contour-like constraints, they enabled the reconstruction of hydrologically consistent reservoir bathymetry in subsequent steps.

B. Bathymetric Reconstruction Results

Bathymetric reconstruction results were evaluated in two phases. First, qualitative analyses were carried out to compare the morphology of the reconstructed bathymetric DEMs. Second, quantitative evaluations were performed using water levels estimated from satellite images excluded from reconstruction. The latter provides an indirect evaluation of the reconstructed DEMs through water-level estimation and is presented in Section IV-C.

Fig. 7 shows the topographic DEM around the Nogok Reservoir. In Fig. 7(a), the interior of the reservoir and the connected upstream and downstream sections (indicated in blue) were overlaid on a topographic DEM. The DEM cells outside of these areas were back-projected to the satellite images as shown in Fig. 7(b). These points contained terrain heights near the reservoir of interest and were used as constraints during the hydrological bathymetric interpolation. They enhanced the structural integrity of the reconstructed bathymetry by ensuring that only hydrologically valid terrain features influenced the interpolation. For the other two reservoirs, the same process was applied to get constraints on nearby terrain heights.

The bathymetric reconstruction was carried out twice, without and with the reference water boundary, whose true elevation values were known. This was to check the applicability of the proposed method of bathymetric reconstruction from time-series water boundaries. Fig. 8 shows the results of Nogok Reservoir, Fig. 9 shows the Maegok Reservoir, and Fig. 10 shows the Pungdo Reservoir.

The top left images in Figs. 8–10 show the original images around the reservoir of interest. The top right images show topographic DEMs. Across all three reservoirs, the topographic DEMs contain unrealistic elevation values on the reservoir surfaces. This is due to the difficulties of precise elevation measurements underneath water bodies.

The bottom left images show bathymetric DEMs constructed without a reference water boundary. The water boundaries extracted from six images were used for bathymetric reconstruction. The resulting DEMs contain bathymetric terrains characterized by smooth slopes and well-formed central depressions, preserving continuity in surface curvature with the adjacent terrain. The bottom right images show

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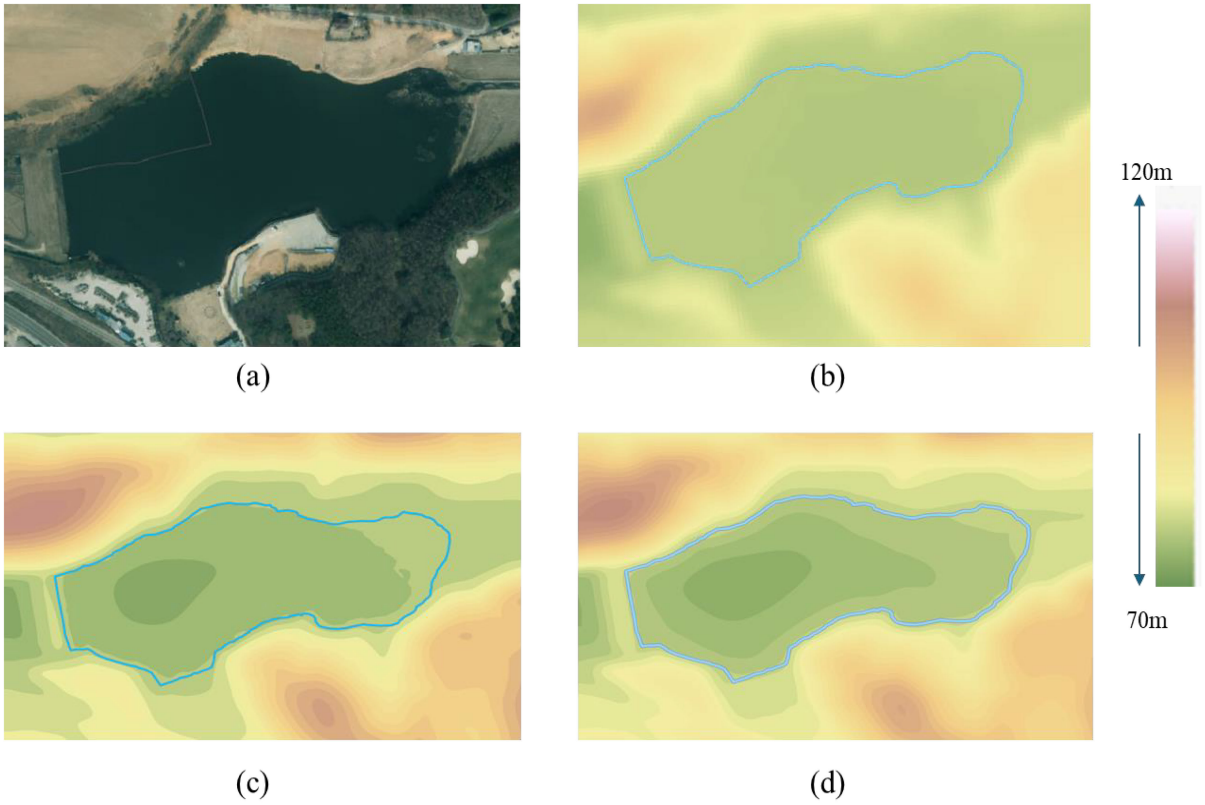


Fig. 8. Nogok reservoir interpolation result. (a) Original image, (b) topographic DEM, (c) bathymetric DEM without reference boundary, and (d) bathymetric DEM with reference boundary.

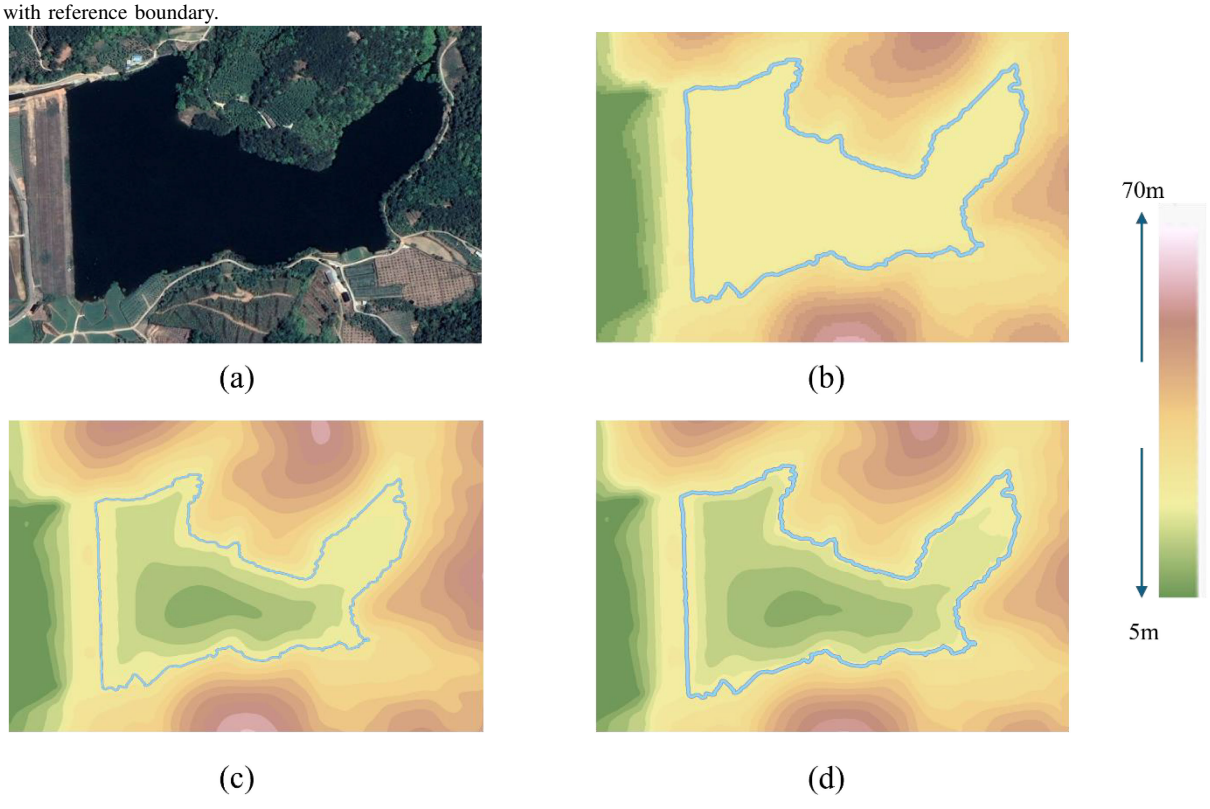


Fig. 9. Maegok reservoir interpolation result. (a) Original image, (b) topographic DEM, (c) bathymetric DEM without reference boundary, and (d) bathymetric DEM with reference boundary.

bathymetric DEMs constructed with one reference water boundary and five nonreference water boundaries. The bathymetric DEMs with one reference water boundary produced more detailed representations of reservoir bathymetry.

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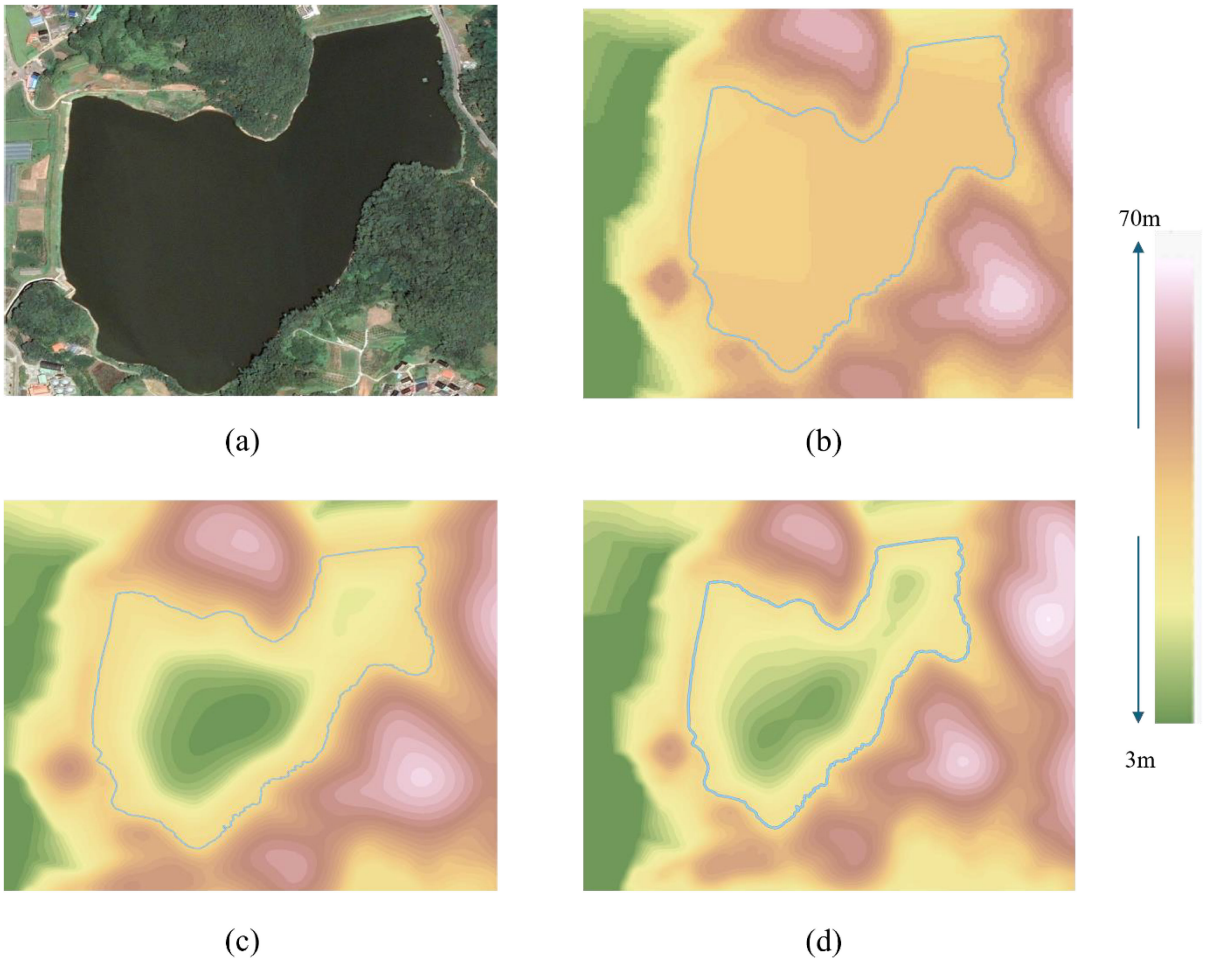


Fig. 10. Pungdo reservoir interpolation result. (a) Original image, (b) topographic DEM, (c) bathymetric DEM without reference boundary, and (d) bathymetric DEM with reference boundary.

Although the reconstruction without reference elevations was able to reproduce the general concave shape of the reservoirs, the reconstruction with reference elevations contained the central depressions and reservoir margins in close agreement with the water boundaries. Nevertheless, the DEMs reconstructed without a reference water boundary showed basin shapes comparable to those reconstructed with a reference boundary. This supports the applicability of the bathymetric reconstruction method proposed here.

The impact of the number of water boundaries on the morphology of bathymetric reconstruction was assessed. Bathymetric interpolation was tested with one, three, and five nonreference water boundaries. In all cases, one reference boundary and the topographic DEMs were provided as constraints.

The bathymetric interpolation treated water boundaries as hierarchical constraints. Accordingly, it was predicted that increasing the number of nonreference boundaries would allow a more precise representation of bathymetric terrain. Figs. 11–13 show the results of bathymetric interpolation using one, three, and five nonreference water boundaries.

Fig. 11 presents the results for Nogok Reservoir, demonstrating that even with one reference and one nonreference water boundary, a central concave structure was partially formed, capturing the basic outline of the reservoir bathymetry. As the number of nonreference water boundaries increased to three and five, the central curvature became progressively more defined, and transitions in water depth were more clearly expressed. The overall terrain morphology and slope patterns remained largely consistent, with only limited visual differences observed.

In contrast, for more topographically complex reservoirs such as Maegok and Pungdo, the improvements in terrain definition were more pronounced. When one reference and one nonreference boundary were used [as shown in Figs. 12(a) and 13(a)], the reservoir interior appeared as a relatively smooth and uniform surface, exhibiting minimal elevation variation and a weak central depression. With one reference and three nonreference boundaries [as shown in Figs. 12(b) and 13(b)], the central concavity became more evident, accompanied by a clearer radial increase in elevation toward the edges. When one reference and five nonreference boundaries were applied [as shown in Figs. 12(c) and 13(c)], the concave structure was more distinctly expressed, and the transitions in water depth were more clearly represented, resulting in DEMs that resemble stepped contour patterns. Overall, this contributed to

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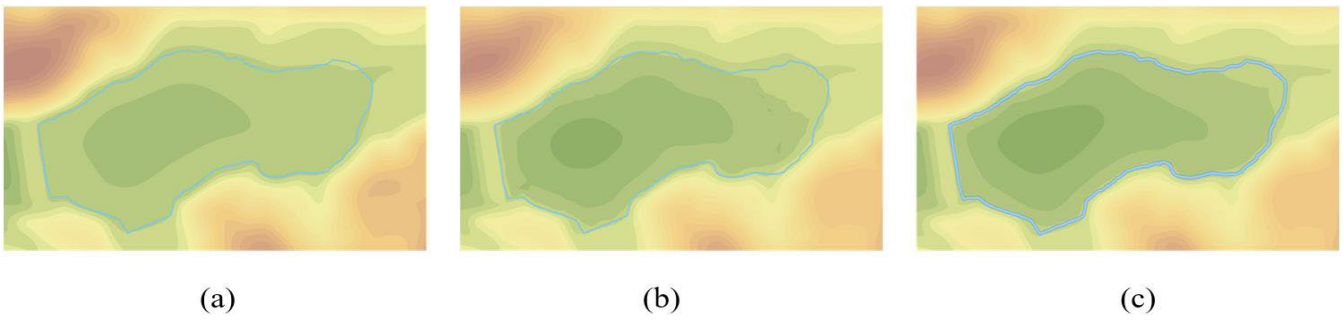


Fig. 11. Nogok reservoir results with (a) one reference and one nonreference boundary, (b) one reference and three nonreference boundaries, and (c) one reference and five nonreference boundaries.

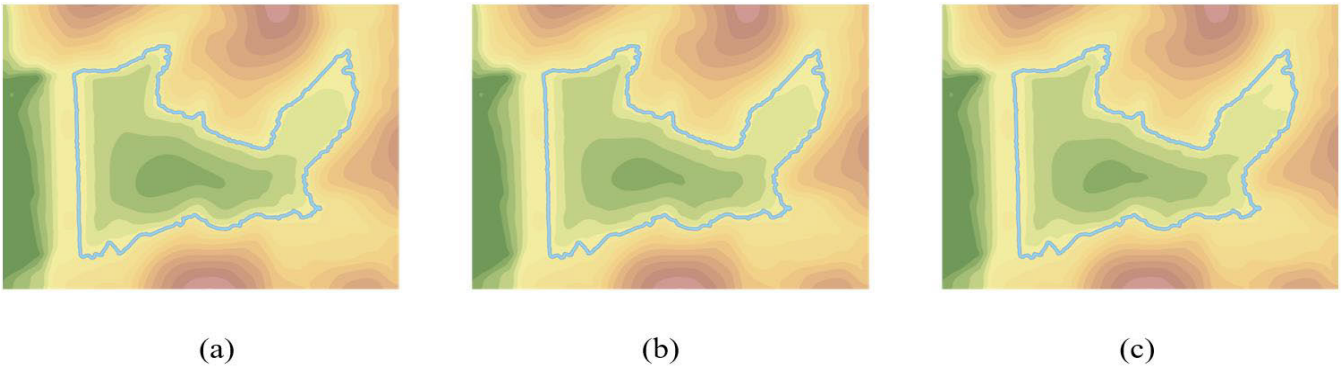


Fig. 12. Maegok reservoir results with (a) one reference and one nonreference boundary, (b) one reference and three nonreference boundaries, and (c) one reference and five nonreference boundaries.

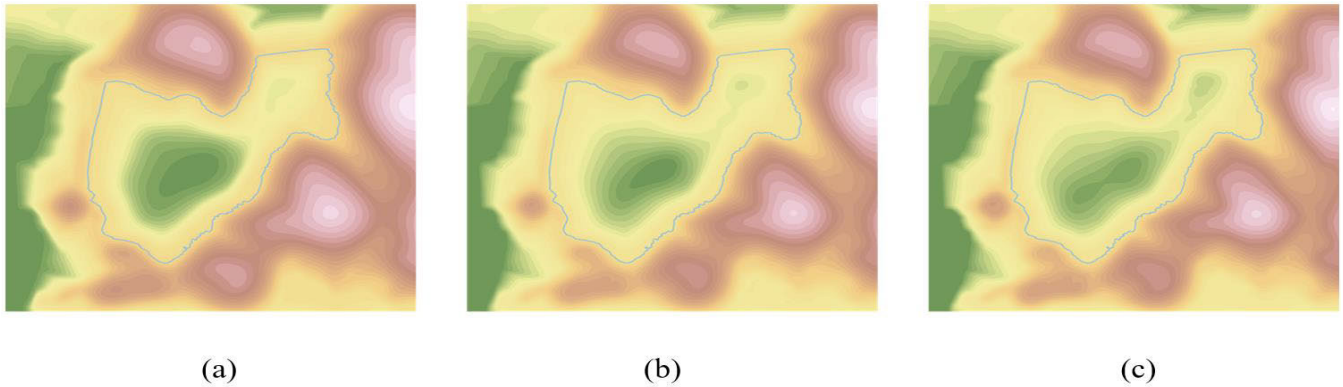


Fig. 13. Pungdo reservoir results with (a) one reference and one nonreference boundary, (b) one reference and three nonreference boundaries, and (c) one reference and five nonreference boundaries.

improvements in representing both curvature and bathymetric gradients.

Additional qualitative analyses of the bathymetric reconstruction results were carried out. The impact of the number of reference boundaries on the bathymetric DEM reconstruction was assessed. The total water boundaries used for the reconstruction were fixed at six. Interpolation was tested with three configurations: one reference and five nonreference water boundaries, three reference and three nonreference water boundaries, and five reference and one nonreference water boundary. Figs. 14–16 show the results from the three configurations.

The results indicated that increasing the number of reference boundaries led to progressively more detailed representation of reservoir bathymetry. This trend was more pronounced in reservoirs with complex geometries (as shown in Figs. 15 and 16) than in relatively simple basins (as shown in Fig. 14). In particular, a comparison of cases (a), (b), and (c) in Figs. 14–16 showed that terrain representation along reservoir margins became clearer as the number of reference elevations increased.

The increased number of reference boundaries produced more detailed bathymetric representations in the visual comparison. Nevertheless, the reconstruction with a single reference boundary retained the general basin structure in the visual comparison. The quantitative evaluation of the reconstructed DEMs through water-level estimation is presented next.

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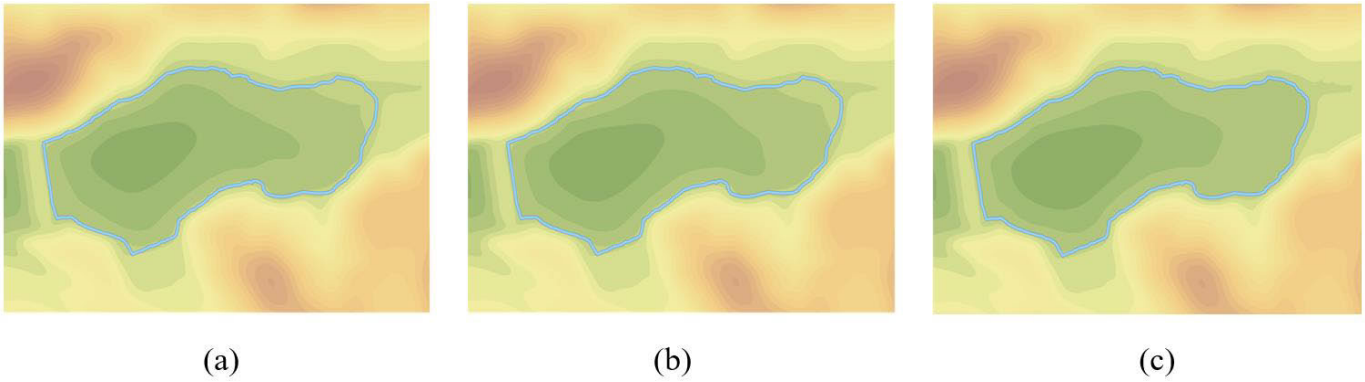


Fig. 14. Nogok reservoir results with (a) one reference and five nonreference boundaries, (b) three reference and three nonreference boundaries, and (c) five reference boundaries and one nonreference boundary.

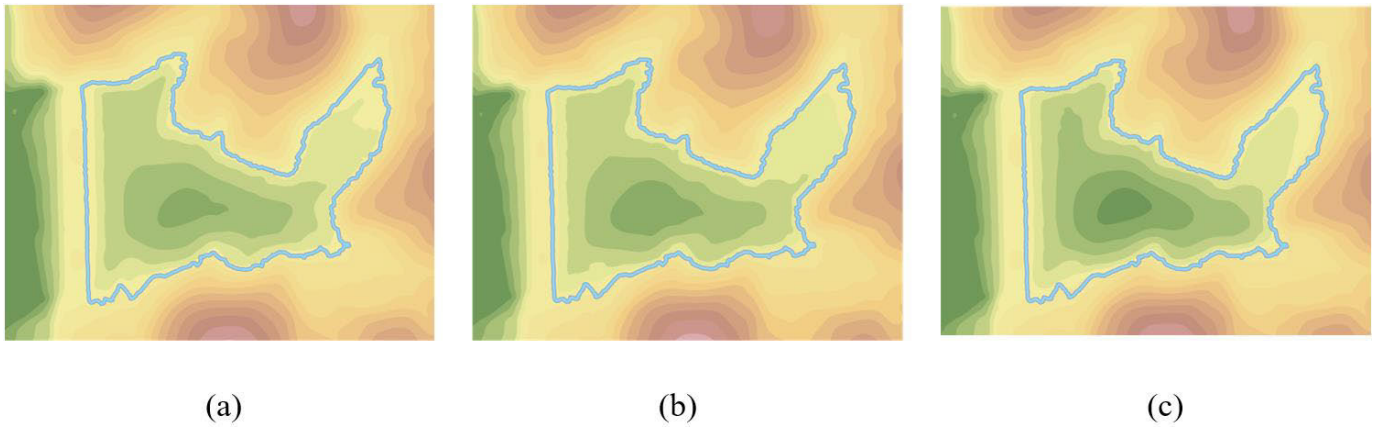


Fig. 15. Maegok reservoir results with (a) one reference and five nonreference boundaries, (b) three reference and three nonreference boundaries, and (c) five reference boundaries and one nonreference boundary.

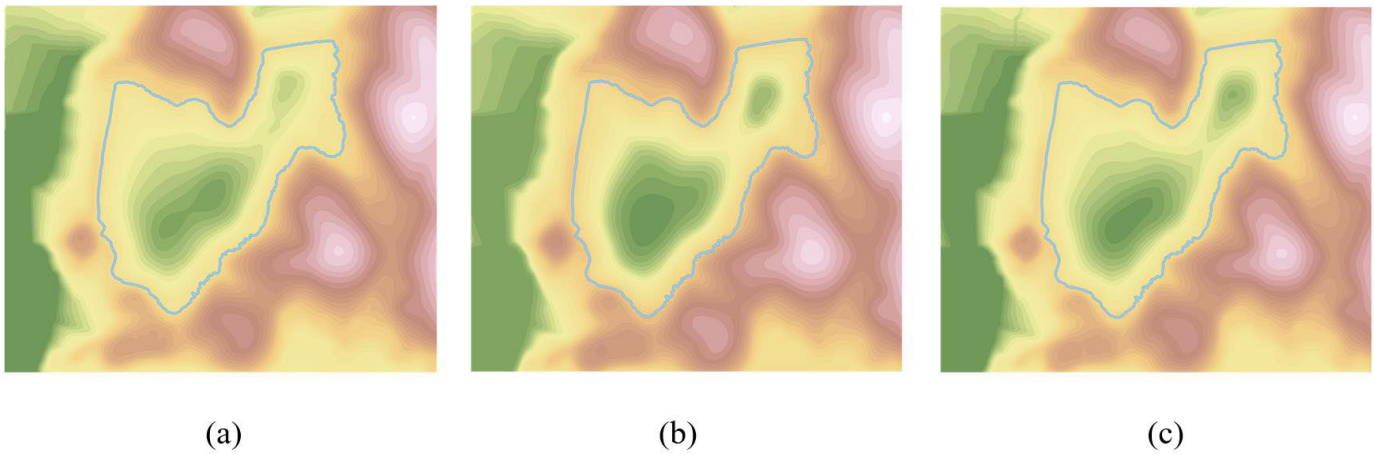


Fig. 16. Pungdo reservoir results with (a) one reference and five nonreference boundaries, (b) three reference and three nonreference boundaries, and (c) five reference boundaries and one nonreference boundary.

C. New Water-Level Estimation Results

For each reservoir, one image was excluded from bathymetric reconstruction and used for water-level estimation. This process was repeated for the six images other than the reference water boundary image. Fig. 17 shows an example for Pungdo Reservoir. The water boundary extracted from the excluded image was overlaid on the reconstructed bathymetric DEM. The elevations of the DEM cells along the water boundary were averaged and set as the estimated water level. The estimated water levels were compared with the corresponding in situ water levels. Table VI shows the reservoir-averaged accuracy statistics for the five boundary configurations. For each configuration, the accuracy statistics were first calculated separately for each reservoir using six validation images, and the values in Table VI were then obtained by averaging the

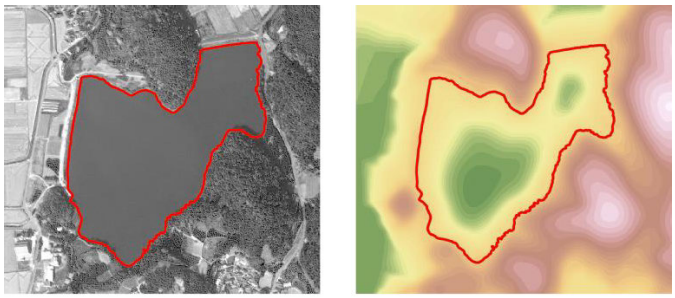
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TABLE VI
WATER-LEVEL ESTIMATION ACCURACY FOR DIFFERENT BOUNDARY CONFIGURATIONS

Number of Reference Boundaries	Number of non-reference Boundaries	Bias (m)	SD (m)	MAE (m)	RMSE (m)	Median AE (m)
1	1	-1.499	0.630	1.572	1.637	1.516
1	3	-0.069	0.874	0.766	0.923	0.857
1	**5**	0.316	0.584	0.609	0.697	0.685
3	3	0.013	0.585	0.505	0.634	0.469
5	1	0.313	0.251	0.434	0.451	0.519

TABLE VII
RESERVOIR-SPECIFIC WATER-LEVEL ESTIMATION ACCURACY USING ONE REFERENCE AND FIVE NONREFERENCE BOUNDARIES

Reservoir	Bias (m)	SD (m)	MAE (m)	RMSE (m)	Median AE (m)
Nogok	0.057	0.751	0.611	0.745	0.571
Maegok	0.495	0.627	0.746	0.799	0.937
Pungdo	0.397	0.374	0.471	0.546	0.548



(a) (b)

Fig. 17. Example of validation using an image excluded from bathymetric reconstruction. (a) Pungdo reservoir satellite image with extracted water boundary. (b) Reconstructed DEM with overlaid water boundary.

three reservoir-specific statistics. Table VII shows the results for each reservoir when one reference water boundary and five nonreference water boundaries were used. The estimation errors were summarized using bias, SD, MAE, RMSE, and median absolute error; all reported in meters.

The first three rows in Table VI correspond to the configurations in Figs. 11–13. These DEMs were reconstructed using one reference boundary and varying numbers of nonreference boundaries. With one reference water boundary, the MAE decreased from 1.572 to 0.766 m and 0.609 m as the number of nonreference water boundaries increased from one to three and five, respectively. The RMSE also decreased from 1.637 to 0.923 m and 0.697 m. These results agree with the qualitative differences shown in Figs. 11–13. The bias changed from −1.499 m with one nonreference boundary to −0.069 and 0.316 m with three and five nonreference boundaries, respectively, indicating that the number of input boundaries affected both the error magnitude and error direction.

The last three rows in Table VI summarize the configurations shown in Figs. 14–16. When six water boundaries were used in total, the MAE decreased from 0.609 to 0.505 m and 0.434 m as the number of reference water boundaries increased from one to three and five, respectively. The RMSE decreased from 0.697 to 0.634 m and 0.451 m. The configurations with three or five reference boundaries were tested to examine the effect of additional reference elevations, although they require more in situ water-level data.

Table VII shows the results for each reservoir under the configuration using one reference water boundary and five nonreference water boundaries. Six water-level estimates were evaluated for each reservoir. The MAE values were 0.611 m for Nogok, 0.746 m for Maegok, and 0.471 m for Pungdo. The results indicate that the method was applicable to the tested reservoirs with different basin geometries. Among the configurations in Table VI, the case using five reference boundaries and one nonreference boundary yielded the lowest reservoir-averaged MAE and RMSE, but it required five measured reference elevations. In contrast, the configuration using one reference boundary and five nonreference boundaries yielded a reservoir-averaged MAE of 0.609 m and an RMSE of 0.697 m while requiring only one measured reference elevation. Therefore, this configuration was selected as the main configuration in this study.

D. Comparison With Representative Methods

To compare the proposed method with relevant representative methods, the same validation procedure described in Section IV-C was used. For each reservoir, each nonreference

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TABLE VIII
COMPARISON WITH REPRESENTATIVE WATER-LEVEL ESTIMATION METHODS

Method	Bias (m)	SD (m)	MAE (m)	RMSE (m)	Median AE (m)
DEM sampling	2.450	1.657	2.467	2.931	3.331
IDW	3.151	1.293	3.151	3.392	3.254
Spline	1.967	1.233	1.967	2.302	2.008
EBK	1.279	1.809	1.567	2.171	0.626
Proposed method	0.316	0.584	0.609	0.697	0.685

image excluded from bathymetric reconstruction was used for water-level estimation, and the estimated water level was compared with the corresponding in situ water level.

The comparison consisted of direct DEM sampling and three interpolation baselines: IDW, spline interpolation, and EBK. In direct DEM sampling, elevations were sampled from the existing topographic DEM along the validation water boundary and averaged without vertical correction. For IDW, spline interpolation, and EBK, elevation surfaces were generated from surrounding topographic DEM cells, and the reference water boundary was assigned the observed water level. The remaining water boundaries were not supplied as interpolation constraints. In the ANUDEM-based reconstruction, these boundaries were incorporated as contour-like constraints, while the reference boundary provided the absolute elevation anchor. The numerical comparison is reported in Table VIII. Direct DEM sampling produced an MAE of 2.467 m and an RMSE of 2.931 m, suggesting that direct boundary sampling from the existing topographic DEM was insufficient for boundary-based water-level estimation. IDW produced the largest MAE and RMSE, with values of 3.151 and 3.392 m, respectively. Spline interpolation reduced the errors to an MAE of 1.967 m and an RMSE of 2.302 m, and EBK further reduced them to 1.567 and 2.171 m.

The proposed method produced the lowest MAE, RMSE, and SD among the relevant representative methods. EBK had a slightly lower median absolute error than the proposed method, 0.626 m compared with 0.685 m, but its SD and RMSE were substantially larger. The improvement can be explained by the incorporation of the remaining time-series water boundaries as contour-like constraints in the reconstruction process. These constraints provided additional geometric information that was not available to the direct DEM sampling or interpolation methods, while the reference water boundary served as the absolute elevation anchor. As a result, the proposed method reduced both the overall error magnitude and error variability under the one-reference water-level condition.

E. Sensitivity Analysis of Uncertainty Sources

Component sensitivity tests were performed for image co-registration, water-boundary extraction, and ANUDEM parameter settings. The tests were applied to the same 18 validation cases used in the accuracy assessment. In each test, the water level estimated after changing a component was compared with the estimate from the adopted setting. The absolute difference between the two estimates was used as the sensitivity value. Therefore, the values in Table IX represent changes in the estimated water level, not errors against the in situ water level.

TABLE IX
SENSITIVITY TESTS FOR UNCERTAINTY SOURCES

Component	Mean absolute change (m)	Maximum absolute change (m)
Image co-registration	0.023	0.091
Water-boundary extraction	0.016	0.065
ANUDEM parameter settings	0.180	0.610

For image co-registration, each validation water boundary was shifted by the reservoir-specific co-registration RMSE in eight directions. For water-boundary extraction, each boundary was shifted by 0.5 and 1.0 pixels in eight directions. For ANUDEM, one parameter group was changed at a time while the output cell size was fixed at 0.55 m. The tested values were maximum iterations of 20 and 30, a roughness penalty of 0.2, a profile-curvature roughness penalty of 0.2, a discretization error factor of 4, and Tolerance 1/Tolerance 2 pairs of 1.0/100 and 5.0/200.

Table IX summarizes the sensitivity test results. The mean absolute changes caused by image co-registration and water-boundary extraction were 0.023 and 0.016 m, respectively. Their maximum absolute changes were 0.091 and 0.065 m. These values were much smaller than the validation RMSE of 0.697 m. ANUDEM parameter settings produced a larger mean absolute change of 0.180 m and a maximum absolute change of 0.610 m. ANUDEM parameter changes produced larger water-level changes than the boundary-position tests.

Uncertainty in in situ water-level measurements and vertical errors in the source topographic DEM were not separately tested because independent error information for these data was not available. Their effects are reflected in the validation errors because the estimated water levels were compared with the corresponding in situ water-level measurements.

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V. Conclusion

This study proposed a method for estimating reservoir water levels from a single satellite image, based on bathymetric reconstruction using time-series water boundaries. The method uses hydrological interpolation in which time-series water boundaries are treated as pseudo-contour constraints and one reference water boundary provides an absolute vertical anchor. This approach reduces the amount of required in situ water-level data while maintaining hydrological consistency in the reconstructed bathymetric surface.

With one reference water boundary and five nonreference water boundaries, the proposed method yielded an MAE of 0.609 m and an RMSE of 0.697 m, calculated from six validation images for each of the three reservoirs. The results indicate that time-series water boundaries can act as effective constraints for bathymetric interpolation and support water-level estimation across reservoirs with different basin geometries. The remaining errors may be affected by spatial resolution, residual geometric co-registration error, boundary extraction quality, ANUDEM parameter settings, and morphological changes in the reservoir. Because the reconstructed bathymetric surface was evaluated indirectly through independent water-level estimation, additional validation using surveyed bathymetric data would further strengthen the assessment.

In summary, this approach provides a feasible method for satellite-based water-level estimation in small- and medium-sized reservoirs with limited in situ measurements.

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